Hyeongchan Ham

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Research Interests

- **Safe robot learning and planning, uncertainty-aware control, Hamilton-Jacobi reachability, world models, and latent-space safety.**

Education

- **Korea Advanced Institute of Science and Technology (KAIST)** — Daejeon, Republic of Korea
  M.S. in Electrical Engineering — *Feb. 2025 – present*
- **Ulsan National Institute of Science and Technology (UNIST)** — Ulsan, Republic of Korea
  B.S. in Electrical and Computer Engineering — *Feb. 2017 – Feb. 2021*
  GPA: 4.01/4.30 (97.10 / 100) ; ***Summa Cum Laude***

Research Experience

- **CIS Lab - KAIST Control Laboratory** — Daejeon, Republic of Korea
  Researcher — *Jun. 2024 – present*
  - **Research (Ongoing): Semantically Safe Navigation Among Movable Obstacles**:
    Developing a latent-space safety framework for Navigation Among Movable Obstacles (NAMO), where semantic failures such as object falling are difficult to model analytically. Integrating learned world representations and Hamilton-Jacobi reachability for safety-aware motion planning.
  - **Research: Safe Motion Planning under Perception Uncertainty**:
    Researched distributionally robust model predictive control algorithms for safe autonomous driving under uncertain neural network perception; uncertainty estimation via evidential deep learning and control via distributionally robust optimization.
- **Agency for Defense Development - AI Autonomy Center** — Daejeon, Republic of Korea
  Research Officer for National Defense — *Jun. 2021 – May. 2024*
  - **Project: AI-enabled UAV Swarm Operations**:
    Developed a real-time algorithm for small object detection and tracking, and deployed it to edge device runtimes. Designed the overall system to enable cooperative perception based on target recognition across multiple agents.
  - **Project: Adaptive Path Planning Based on Situational Awareness and Dynamic Model Learning**:
    Constructed a real-time obstacle perception system and developed a 3D object detection algorithm based on point cloud streams operating on the ROS2 platform; data logging, labeling, streaming, and visualization.
- **OMNIA - UNIST System Laboratory** — Ulsan, Republic of Korea
  Research Intern — *Jul. 2020 – Feb. 2021*
  - **Project: Continual Learning**:
    Developed an efficient continual learning dataloader system for the replay buffer, focusing on maximizing data replay stream throughput.

Work Experience

- **Research Officer for National Defense** — Republic of Korea
  First Lieutenant, Republic of Korea Air Force — *Jun. 2021 – May. 2024*
  - Selected as one of only 20 officers nationwide to conduct full-time science and technology research for national defense.

Peer-reviewed Publications

1. **S2-NAMO: Semantically Safe Navigation Among Movable Obstacles**
   - Hyeongchan Ham, Boseok Kim, Minwoo Song, Heejin Ahn
   - Workshop on Rethinking What It Means to be "Safe" for Generalist Robots @ RSS 2026 (Spotlight)

2. **DRO-EDL-MPC: Evidential Deep Learning-Based Distributionally Robust Model Predictive Control for Safe Autonomous Driving**
   - Hyeongchan Ham, Heejin Ahn
   - IEEE Robotics and Automation Letters (RA-L), 2026. Presented at ICRA 2026.
   - Workshop on Safe and Robust Robot Learning for Operation in the Real World @ CoRL 2025
   - https://arxiv.org/pdf/2507.05710

3. **Probabilistic Recursively Feasible Model Predictive Control Under Uncertain Environments**
   - Hyeontae Sung, Hyeongchan Ham, Junyoung Park, Kai Ren, and Heejin Ahn
   - In 23rd IFAC World Congress, 2026
   - https://arxiv.org/pdf/2605.19015v1

4. **Multi-Class Multi-Object Tracking in Aerial Images Using Uncertainty Estimation**
   - Hyeongchan Ham, Junwon Seo, Junhee Kim, Chungsu Jang
   - Korean Journal of Remote Sensing, 2024, 40(1), 115-122.

Domestic Publications

1. **A Survey on Safe Model Predictive Control and Reinforcement Learning**
   - Hyeongchan Ham, Heejin Ahn
   - Journal of Institute of Control, Robotics and Systems 2025, 31(2), 87-97.

Scholarship

- Hyundai CMK Scholarship (Future Industries Scholarship) *2025 – present*
- Korea National Scholarship of Excellence (Science and Engineering) *2019 – 2020*
- UNIST Academic Performance Scholarship (Full Scholarship) *2017 – 2020*

Teaching Experience

- **Mentor, CSE winter school, UNIST** *Dec. 2020 – Feb. 2021*
  Linux Tutoring Program of the School of Computer Engineering
- **Teaching Assistant, Data Structure, UNIST** *Sep. 2020 – Dec. 2020*
  (English) C++ / Data Structure

Skills

- **Programming Languages:** Python, C, C++, Java, JavaScript
- **Technologies:** PyTorch, TensorFlow, JAX, ONNX, TensorRT, Docker, Linux, MLFlow, DeepStream, ROS2